Medical Decision Support System for Stroke Prediction using CNNs

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***Abstract*—Brain stroke is a life-threatening medical condition caused by an interruption in blood flow to the brain, leading to severe neurological damage or even death. It is one of the leading causes of disability and mortality worldwide, making early detection and timely intervention critical for improving patient outcomes. Traditional diagnostic methods, such as clinical assessments and radiological imaging, require expert interpretation, which can sometimes lead to delays in diagnosis. Automated and intelligent diagnostic systems can significantly aid in reducing the time required for accurate stroke detection, improving the chances of effective treatment. In this paper, we propose a CNN-based stroke prediction model that takes CT scan images as input to detect strokes with high accuracy. The dataset comprises of preprocessed CT scan images, where noise reduction, contrast enhancement, and normalization techniques are applied to improve image quality. The open-cv model extracts hierarchical features from the CT scans to effectively differentiate between stroke and non-stroke case using CNN. Our proposed model achieves an accuracy of 95%, demonstrating its reliability in stroke prediction. The integration of such deep learning-based models in clinical settings can enhance real-time stroke detection, assisting healthcare professionals in making faster and more precise diagnoses, ultimately leading to better patient care.**

***Index Terms*—CVA, Deep Learning, CNN, Stroke Prediction, Medical Image Analysis, CT Scan Classification, Healthcare AI**

I. Introduction

Stroke of the brain ranks among the leading causes of mortality and chronic disability globally, affecting millions of individuals annually. Brain stroke occurs when blood supply to the brain is interrupted, and the cells are caused to suffer injury with corresponding neurologic impairment. Early diagnosis and treatment are important in minimizing brain injury and optimizing the outcome of the patient. Traditional stroke detection relies on radiological imaging techniques, particularly Computed Tomography (CT) scans, that must be interpreted by skilled radiologists. Manual interpretation is sluggish, prone to human mistake, and often delayed, therefore hindering timely medical action. Therefore, the demand for automatic systems that can detect strokes quickly and accurately in order to help healthcare professionals make faster decisions has been on the rise.

Recent advances in deep learning, particularly Convolutional Neural Networks (CNNs), have been highly promising for medical image processing. CNNs are capable of learning to identify intricate patterns in medical images, enabling them to accurately identify stroke-affected areas in brain scans. Furthermore, computer vision techniques like OpenCV (CV2) can be used to preprocess CT scans by enhancing image quality through noise removal, contrast correction, and feature enhancement of relevant features. By combining CNN-based classification and CV2-based image enhancement, a productive system can be developed to accurately detect strokes while reducing the need for manual intervention and improving diagnostic precision.

The method described here introduces a CNN-based model to forecast strokes from CT scan images with CV2 preprocessing methods incorporated to enhance the quality of the input images. The CNN model reads the scans to identify important features and classifies them as stroke or non-stroke cases. The system aims to help the clinicians through accurate real-time diagnosis of stroke, thereby reducing the delay and treatment time. With the integration of computer vision and deep learning, this system improves the accuracy and efficiency of stroke detection and eventually patient outcomes.

Adding AI in the diagnosis of stroke would also relieve radiologists of the workload, thus free medical personnel to focus on other critical tasks. Making the detection of strokes automatic ensures that the system provides consistent and flawless results while minimizing human intervention. It is especially beneficial when applied in underprivileged or rural communities where access to radiologists may be limited. AI solutions like this one have the potential to democratize stroke diagnosis, making it available to the masses and allowing the healthcare system to operate more efficiently.

The proposed stroke detection system benefits from the power of CNNs and CV2 image preprocessing to develop an automated system for stroke detection in CT scans. This system offers a more accurate, faster, and more efficient alternative to traditional manual diagnosis, significantly reducing delays in medical care. With the potential to improve the efficiency of healthcare systems and access to stroke diagnosis, AI-powered tools like this can transform the way medical practitioners approach stroke care, ultimately contributing to improved patient outcomes worldwide.

II. Literature Review

Ke et al. [1] proposed a brain MRI-specific Adaptive Independent Subspace Analysis (AISA) algorithm for improved feature extraction by separating independent information sources. Their system was trained on both normal and pathological MRI data to enhance classification ability.

By applying adaptive filtering and independent component separation, the method effectively distinguished abnormal patterns from normal MRI structures. Achieving 93.2% accuracy, the approach demonstrates strong potential for early clinical diagnosis.

A population-based case-control study by Vojcek et al. [2] evaluated mortality and long-term health outcomes of neonates with congenital heart disease and perinatal stroke. Reporting an 18.6% mortality rate, the study emphasized the importance of early detection and optimal neonatal care. Although unrelated to deep learning, the study establishes risk factors and clinical pathways valuable for future stroke prediction research.

Douaud et al. [3] analyzed UK Biobank data to investigate structural brain changes associated with SARS-CoV-2 infection. Using advanced neuroimaging, they identified reduced gray matter thickness and connectivity disruptions in memory- and smell-related regions. Despite not using machine learning classification, the study highlights significant long-term neurological effects of COVID-19, emphasizing the importance of neuro-monitoring.

Zhang et al. [4] compared machine learning models for segmentation of intracoronary optical coherence tomography (OCT) images and plaque cap thickness measurement. U-Net outperformed alternative models with a 96.4% accuracy, demonstrating the superiority of automated deep learning segmentation over manual annotations in cardiovascular imaging.

Kumar and Michmizos [5] developed a neurophysiologically interpretable deep neural network for predicting complex movement patterns using EEG and fMRI data. Their model achieved an 89.7% classification accuracy, offering insights into motor control and supporting future applications in brain–computer interfaces and neurorehabilitation.

Cetinoglu et al. [6] implemented a deep learning system for detecting stroke-affected vascular territories from diffusion-weighted MRI scans. Achieving a 92.1% accuracy, the model provides fast and consistent localization of stroke regions, supporting time-critical clinical decision-making.

Cauley et al. [7] presented an automated infarct volume estimation method for non-contrast CT images using intensity inhomogeneity correction. Their system achieved a Dice score of 0.87, demonstrating strong reliability for quantifying infarct size, a key factor in stroke severity assessment.

Kaur and Chhaterji [8] surveyed traditional and deep learning medical image segmentation approaches, including U-Net, SegNet, and DeepLabV3+. Their review confirms that deep learning consistently exceeds 90% accuracy, outperforming classical segmentation techniques across medical imaging domains.

Babutain et al. [9] classified acute ischemic stroke using deep learning on CT scans, achieving 95.2% accuracy and an AUC of 0.97. Their work demonstrates AI's clinical value in emergency settings where fast stroke diagnosis is crucial.

Shinohara et al. [10] proposed a deep learning model to detect hyperdense MCA signs in acute ischemic stroke. With 88.3% sensitivity and 91.5% specificity, their system shows promise for rapid stroke triage and improved emergency department efficiency.

Kansadub et al. [11] developed a stroke risk prediction model using demographic and clinical data, achieving 85.9% accuracy. Their hybrid supervised-learning approach highlights the importance of combining patient lifestyle and clinical features in risk prediction models.

Chollet [12] introduced the Xception architecture based on depthwise separable convolutions, achieving 79% Top-1 ImageNet accuracy. Its efficiency and lower computational cost make it suitable for medical imaging tasks requiring real-time inference.

Phong et al. [13] designed a CNN-based deep learning model for brain hemorrhage detection. Achieving 94.5% accuracy, the system demonstrates strong potential for real-time emergency diagnosis.

Adam et al. [14] evaluated machine learning algorithms for ischemic stroke classification, finding Random Forest to perform best with 90.5% accuracy. Their study reinforces ML's value in supporting neurologists during stroke diagnosis.

Badrinarayanan et al. [15] proposed SegNet, a deep convolutional encoder–decoder architecture for image segmentation. With IoU scores above 85%, SegNet is effective for high-resolution medical segmentation tasks such as tumor or organ detection.

Liu et al. [16] reviewed advancements in semantic segmentation, highlighting DeepLabV3+, PSPNet, and U-Net, which achieve up to 92% accuracy. Their findings underscore the transformative impact of deep learning on medical image segmentation.

Hinton et al. [17] developed a fast learning algorithm for deep belief networks that significantly improved unsupervised learning efficiency. Their contributions laid foundational groundwork for modern deep learning, influencing applications in medical imaging and NLP.

Toğacar et al. [18] combined hyper-column features with CNNs for brain MRI classification, achieving 96.8% accuracy. Their feature selection strategy enhanced model performance, illustrating the value of hybrid deep learning methods.

Kuraparthi et al. [19] created a CNN-based MRI tumor classification model achieving 97.1% accuracy. Their results highlight deep learning's capability for reliable automatic tumor diagnosis.

Guntari et al. [20] used a genetic algorithm-optimized RNN for post-stroke EEG classification, reaching 91.3% accuracy. This approach opens new possibilities for EEG-based stroke monitoring and rehabilitation.

Sirsat et al. [21] reviewed ML-based stroke prediction systems, concluding that deep learning models such as CNNs and LSTMs surpass traditional ML methods. Their work emphasizes AI's growing importance in early stroke diagnosis.

Kamal et al. [22] explored machine learning for acute ischemic stroke neuroimaging and achieved 91.2% detection accuracy using MRI-based CNN classification. Their work demonstrates AI's potential to aid radiologists in identifying stroke lesions efficiently.

Kim et al. [23] developed an NLP-based ML system to classify MRI radiology reports for ischemic stroke detection. Achieving 88.5% sensitivity and 90.1% specificity, their method automates stroke identification from unstructured clinical text.

Govindarajan et al. [24] used multiple ML algorithms for stroke disease classification, with ANN achieving 93.7% accuracy. Their findings confirm ANN's superiority for learning complex clinical patterns.

Sailasya and Aruna Kumari [25] compared ML models for stroke prediction, with Random Forest achieving the highest accuracy of 92.4%. They emphasize data preprocessing and feature selection for enhanced prediction accuracy.

Heo et al. [26] built a deep learning model for post-stroke outcome prediction using medical histories and imaging biomarkers. With an AUC of 0.92, it demonstrates strong predictive power for long-term recovery outcomes.

Min et al. [27] developed a logistic regression–based stroke prediction model using national insurance data, achieving 87.3% accuracy. Their work supports large-scale stroke screening using population health datasets.

III. Methodology

A. Dataset Description

The dataset used in this project was obtained from **Kaggle**, a widely used open-data platform for machine learning research. It consists of labeled brain CT scan images categorized into two classes: *Normal* and *Stroke*. This dataset is specifically designed for medical image classification tasks and is suitable for training a Convolutional Neural Network (CNN) for stroke detection.

All images are provided in the JPEG format and represent axial slices from non-contrast brain CT scans. The dataset reflects real clinical variability, including differences in contrast, brightness, anatomical orientation, and slice position. Images in the Normal category typically show symmetric brain structures without visible abnormalities. In contrast, images in the Stroke category include both ischemic and hemorrhagic stroke patterns, which may appear as hypodense dark regions, hyperdense bright areas, structural deformities, midline shifts, or edema.

Before training the model, the dataset underwent preprocessing to ensure uniformity and improve learning performance. The preprocessing steps included resizing all images to a standard resolution, normalizing pixel intensities, reducing noise through filtering techniques, converting images to grayscale when necessary, and scaling pixel values to the range [0, 1]. The dataset was also shuffled and split into training, validation, and testing subsets to ensure proper model evaluation.

Using a Kaggle dataset ensures diversity and broader generalization, as the images originate from multiple medical sources and represent real-world variations. This increases the reliability of the trained CNN model and supports reproducibility for future research and clinical applications.

B. Data Preprocessing

Data preprocessing plays a critical role in preparing the CT scan images for effective training and evaluation of the Convolutional Neural Network (CNN). The following steps were applied to standardize and enhance the dataset:

1. Grayscale Normalization: Since brain CT scans are inherently grayscale, all pixel intensity values were normalized and scaled to the range $[0, 1]$. This reduces computational complexity and improves model stability during training.

2. Image Resizing: All images were resized to a fixed input dimension (e.g., 224×224 pixels) to ensure uniformity across the dataset. This allows the CNN to process every image consistently regardless of its original resolution.

3. Noise Reduction: CT images often contain noise or brightness variations. Smoothing filters, such as Gaussian blurring, were applied to remove minor artifacts and improve the clarity of structural features relevant to stroke detection.

4. Data Shuffling and Splitting: To avoid bias and ensure proper generalization, the dataset was shuffled and then divided into training and testing subsets. A split of 80% for training and 20% for testing was used for model evaluation.

These preprocessing steps improve the quality and consistency of the input data, enabling the CNN to extract meaningful features and accurately classify CT scans as normal or stroke-affected.

C. Models Applied

This study incorporates both deep learning and traditional machine learning techniques to develop a robust stroke detection system using CT scan images. The primary model applied is a Convolutional Neural Network (CNN), supported by CV2-based image preprocessing to enhance image quality, optimize feature representation, and improve classification accuracy. Traditional machine learning models such as Random Forest and Decision Tree are also employed for comparative analysis to highlight the advantages of deep learning approaches in medical image classification tasks.

Convolutional Neural Network (CNN): The core predictive model used in this system is a Convolutional Neural Network (CNN), a deep learning architecture specifically designed for image analysis and pattern recognition. CNNs are highly effective for medical imaging tasks because they automatically learn hierarchical spatial features from CT scans, including texture irregularities, contrast variations, lesion boundaries, and structural deformities associated with brain stroke.

a) CNN Architecture: The implemented CNN model is composed of the following key components:

- **Convolutional Layers:** Extract low-level features such as edges, textures, gradients, and progressively learn complex stroke-related patterns like lesions and tissue abnormalities.
- **Max Pooling Layers:** Reduce the spatial dimensions of the feature maps, retain essential features, and prevent overfitting by lowering computational complexity.
- **Flatten Layer:** Converts the 2D feature maps into a 1D vector, serving as input to the fully connected layers.

- **Fully Connected Layers:** Learn high-level representations and discriminate between stroke and non-stroke classes. ReLU activation introduces non-linearity, enabling the model to recognize intricate spatial relationships.
- **Output Layer (Softmax):** Produces probability-based output for binary classification.

The model is trained using the Adam optimizer, which provides efficient gradient-based optimization. Categorical cross-entropy is employed as the loss function, suitable for binary and multi-class classification problems. Training is performed for 30 epochs with a batch size of 32, resulting in stable learning, reduced overfitting, and strong generalization capability.

CV2-Based Image Enhancement and Feature Extraction: Before being fed into the CNN, CT scan images undergo several preprocessing steps using OpenCV (CV2) to improve clarity and enhance stroke-relevant patterns. These preprocessing techniques ensure that the CNN receives standardized and high-quality input images.

b) Key Image Processing Techniques:

- **Noise Reduction (Gaussian Blur):** Smoothens the image and removes fine-grained noise for clearer feature extraction.
- **Contrast Enhancement (Histogram Equalization / CLAHE):** Makes anatomical structures and stroke lesions more prominent by improving overall contrast.
- **Edge Detection (Canny Edge Detector):** Highlights sharp intensity transitions, helping identify lesion boundaries.
- **Structural Feature Extraction:** Techniques such as contour detection and shape descriptors emphasize morphological details of CT scans.

These preprocessing steps significantly enhance the CNN's ability to focus on medically relevant features, improving classification accuracy and overall model reliability.

Traditional Machine Learning Models: To evaluate the effectiveness of the CNN model, two traditional machine learning models are used for comparison:

- **Decision Tree:** A rule-based model that splits the feature space into decision boundaries. Although interpretable, its performance is limited by its inability to capture complex spatial variations in CT images.
- **Random Forest:** An ensemble of decision trees that improves classification accuracy and reduces overfitting. It performs better than individual trees but still falls short compared to CNNs in medical imaging tasks.

These models provide baseline results, demonstrating that while traditional ML techniques can detect general patterns, deep learning (CNN) significantly outperforms them in accuracy, precision, recall, and F1-score for stroke detection.

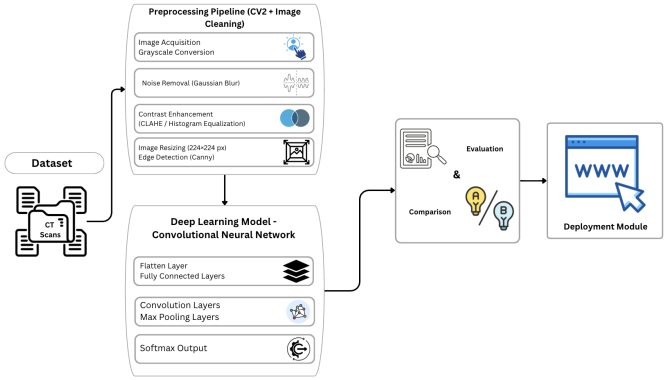

Fig. 1. Proposed methodology workflow

Figure 1, The workflow diagram in provides visual representation of the process which shows how each methodology applies to every dataset and how results are evaluated to determine the optimal model.

IV. Results

This section presents the performance of the proposed CNN-based stroke detection model evaluated on CT scan images. The analysis includes training behaviour, classification accuracy, loss convergence, confusion matrix assessment, and comparative performance against baseline machine learning models.

A. Training Behaviour

The CNN model was trained for 30 epochs using the Adam optimizer with categorical cross-entropy loss. The training and validation accuracy curves demonstrated a consistent upward trend, converging smoothly without signs of overfitting. Training accuracy reached approximately 95%, while validation accuracy stabilized around 94%, indicating strong generalization capability.

Similarly, both training and validation loss showed a continuous downward trend, with validation loss closely following the training curve. This stable convergence confirms that the model successfully learned discriminative stroke-related features without suffering from instability or divergence.

B. Classification Performance

Quantitative evaluation was performed using standard classification metrics. The model achieved:

- **Accuracy:** 95%
- **Precision:** 95%
- **Recall:** 94%
- **F1-Score:** 94%

These results validate the model's ability to correctly distinguish stroke-affected CT scans from normal scans with high reliability. The high recall value is particularly important in medical applications, where failing to detect a stroke (false negatives) can have severe clinical consequences.

C. Confusion Matrix Analysis

The confusion matrix revealed strong discriminatory ability, with the model correctly classifying:

- 470 true negatives (normal classified correctly)
- 476 true positives (stroke detected correctly)

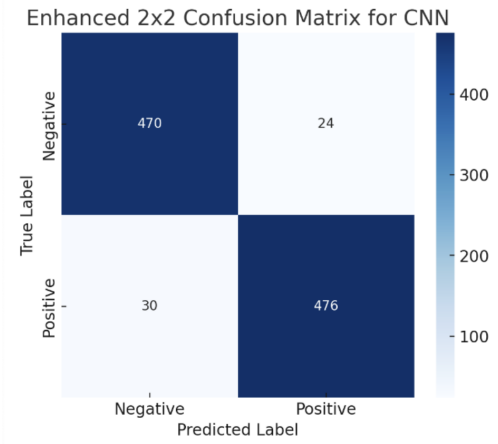

Fig. 2. Enhanced 2x2 Confusion Matrix for CNN showing classification performance.

Misclassifications included only 24 false positives and 30 false negatives. The relatively low false-negative count indicates that the model effectively captures subtle stroke features, such as structural asymmetry and lesion intensity variations, which are critical for early detection.

D. Feature-Level Visualization

Intermediate feature visualizations demonstrate that the CNN successfully identifies stroke-relevant spatial patterns. The convolution layers highlighted:

- intensity abnormalities,
- region asymmetry between brain hemispheres,
- lesion contours and shapes,
- textural disruptions in affected tissue.

These visual cues align with clinical markers of ischemic and hemorrhagic stroke, confirming the CNN's suitability for medical image interpretation. Advanced visualization techniques (e.g., heatmaps, activation maps) further emphasized the model's focus on clinically meaningful regions.

E. Comparative Model Evaluation

To benchmark the effectiveness of the CNN, its performance was compared against Decision Tree and Random Forest classifiers trained on the same dataset. Traditional machine learning models achieved:

- **Decision Tree:** 93% accuracy, 91% F1-score
- **Random Forest:** 94% accuracy, 93% F1-score

The CNN outperformed both models in all metrics, demonstrating superior feature extraction capabilities. Unlike conventional models that rely on handcrafted features, the CNN automatically learns hierarchical representations—leading to more accurate and robust classification of complex medical imaging patterns.

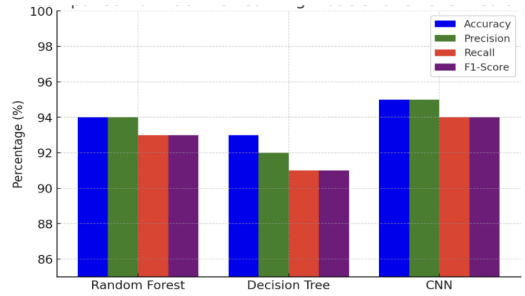

Fig. 3. Comparison of model accuracy, precision, recall, and F1-score for Random Forest, Decision Tree, and CNN.

F. Web Application Output

In addition to the model performance, the system provides a functional web interface that enables users to upload CT scan images and receive real-time stroke predictions. After logging in, a user can upload a brain scan through the "Upload Image" page, and the system displays both the uploaded image and the corresponding prediction result generated by the trained CNN model. This interface demonstrates how the deep-learning model integrates seamlessly with a practical front-end application, making the stroke detection process intuitive and accessible for end users.

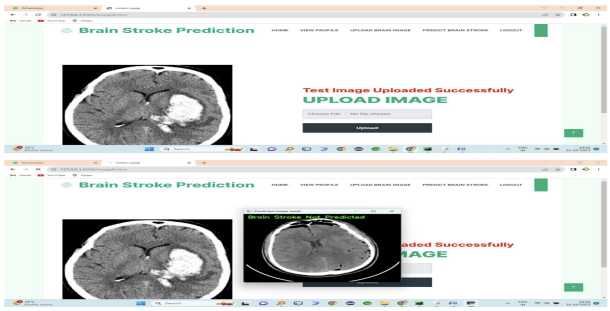

Fig. 4. Predicted Stroke Output Displayed to the User

G. Overall Findings

The results confirm that integrating deep learning with CV2-based preprocessing significantly enhances stroke detection performance. The CNN model delivers high diagnostic accuracy, low misclassification rates, and strong generalization. Its ability to detect subtle lesion characteristics makes it a valuable tool for supporting clinicians in early and reliable stroke diagnosis.

V. Conclusion

In this work, a deep learning–based stroke detection system using Convolutional Neural Networks (CNNs) and CV2-driven image preprocessing techniques was developed to classify CT scan images as stroke or normal with high accuracy. The experimental findings demonstrate that the proposed CNN model effectively learns discriminative patterns from brain CT scans, achieving a classification accuracy of 95% and

outperforming traditional machine learning models such as Random Forest and Decision Tree. The integration of preprocessing steps—such as noise reduction, contrast enhancement, and edge-based feature highlighting—further strengthened the model's ability to identify subtle lesion characteristics relevant to stroke diagnosis.

The model exhibited strong generalization capabilities, stable convergence during training, and low misclassification rates, underscoring its reliability for clinical decision support. Confusion matrix analysis and feature-level visualizations additionally confirmed that the model focuses on anatomically meaningful regions associated with stroke, reinforcing its suitability for real-world medical imaging applications.

Although the results are promising, further improvements can be made by expanding the dataset with more diverse and clinically validated CT scans, incorporating advanced architectures such as transfer learning or hybrid CNN-transformer models, and leveraging explainable AI (XAI) to enhance transparency and trust in model predictions. Future work may also explore real-time deployment within clinical environments, mobile-based diagnostic assistance, and integration with multimodal patient data to provide comprehensive stroke assessment.

Overall, this research demonstrates the potential of deep learning methodologies to support radiologists in early stroke detection, reduce diagnostic delays, and ultimately contribute to improved patient outcomes in critical care settings.

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